Rafael D. De Paz
Independent Researcher
https://logos.pub

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Editorial Board
Communications in Mathematical Physics
Springer

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

"Global Regularity of the 3D Incompressible Navier-Stokes Equations on a Discrete Bounded Lattice,"

for consideration for publication in *Communications in Mathematical Physics.*

Summary. The paper establishes the global-in-time existence, uniqueness, and regularity of solutions to the three-dimensional incompressible Navier-Stokes equations formulated on a discrete periodic lattice $\Lambda_h \subset h\mathbb{Z}^3$ with mesh spacing $h > 0$. The central result is that the finite-dimensional character of the discrete lattice provides a Sobolev embedding $\ell^2(\Lambda_h) \hookrightarrow \ell^\infty(\Lambda_h)$ that is absent in the continuum, yielding uniform *a priori* bounds on the velocity field and precluding finite-time blow-up.

Relation to the Millennium Problem. The paper directly addresses the Navier-Stokes existence and smoothness problem as posed by the Clay Mathematics Institute. We argue, on both physical and mathematical grounds, that the discrete formulation represents the physically correct setting (owing to Planck-scale and molecular-scale discreteness of physical spacetime and matter), and that the continuum formulation $h = 0$ is a mathematical idealization. The paper includes:

- Rigorous proofs of global existence, uniqueness, and energy dissipation on the discrete lattice.
- A formal convergence theorem showing that the discrete solutions converge to Leray-Hopf weak solutions as $h \to 0$.
- A detailed discussion of the "different problem" objection, addressed at the physical, mathematical, and epistemological levels.
- A comparison with Tao's (2016) blow-up obstruction for averaged Navier-Stokes equations.

Novelty and Significance. While discrete Navier-Stokes regularity is well understood in computational fluid dynamics, the present paper is distinguished by its explicit framing as a resolution of the Millennium Problem and its systematic physical argumentation for the primacy of the discrete formulation. The molecular discreteness argument (Section 5.2.2), in particular, requires no assumptions about quantum gravity and provides a self-contained physical justification accessible to the broader mathematical physics community.

Suggested Reviewers. I suggest the following experts as potential referees:

1. **Prof. Terence Tao** (UCLA) — Leading expert on Navier-Stokes regularity and author of the averaged blow-up result discussed in the paper.
2. **Prof. Peter Constantin** (Princeton) — Expert in the analysis of fluid equations and energy methods.
3. **Prof. Carlo Rovelli** (Aix-Marseille) — Leading figure in loop quantum gravity and the discreteness of spacetime.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm that I have no conflicts of interest to declare.

I look forward to your consideration.

Respectfully submitted,

Rafael D. De Paz
Independent Researcher
hi@logos.pub